Questions

Q1.

Use the binomial series to find the expansion of

$$\frac{1}{(2+5x)^3} \qquad |x| < \frac{2}{5}$$

in ascending powers of x, up to and including the term in x^3

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Give each coefficient as a fraction in its simplest form.

(Total for question = 6 marks)

(Q01 WMA14/01, Specimen papers)

Q2.

(a) Use the binomial expansion to expand

$$(4-5x)^{-\frac{1}{2}} \qquad |x| < \frac{4}{5}$$

in ascending powers of x, up to and including the term in x^2 giving each coefficient as a fully simplified fraction.

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(4)

$$f(x) = \frac{2+kx}{\sqrt{4-5x}} \qquad \text{where } k \text{ is a constant and } |x| < \frac{4}{5}$$

Given that the series expansion of f(x), in ascending powers of x, is

$$1 + \frac{3}{10}x + mx^2 + \ldots \qquad \text{where } m \text{ is a constant}$$

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(b) find the value of k,

(2)

(c) find the value of m.

(2)

(Total for question = 6 marks)

(Q02 WMA14/01, Oct 2020)

Q3.

(a) Find the first 4 terms, in ascending powers of x, of the binomial expansion of

$$\left(\frac{1}{4} - 5x\right)^{\frac{1}{2}} \quad |x| < \frac{1}{20}$$

giving each coefficient in its simplest form.

(5)

By substituting $x = \frac{1}{100}$ into the answer for (a),

(b) find an approximation for $\sqrt{5}$

Give your answer in the form $\frac{a}{b}$ where a and b are integers to be found.

(2)

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(Total for question = 7 marks)

(Q01 WMA14/01, Jan 2021)

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Q4.

Given that k is a constant and the binomial expansion of

$$\sqrt{1+kx} \qquad |kx| < 1$$

in ascending powers of x up to the term in x^3 is

$$1 + \frac{1}{8}x + Ax^2 + Bx^3$$

(a) (i) find the value of k,

(ii) find the value of the constant A and the constant B.

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(5)

(b) Use the expansion to find an approximate value to $\sqrt{1.15}$

Show your working and give your answer to 6 decimal places.

(2)

(Total for question = 7 marks)

(**Q01 WMA14/01, June 2021**)

Q5.

$$f(x) = \sqrt{1-4x^2} \qquad |x| < \frac{1}{2}$$

(a) Find, in ascending powers of x, the first four non-zero terms of the binomial expansion of f(x) . Give each coefficient in simplest form.

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(4)

(b) By substituting $x = \frac{1}{4}$ into the binomial expansion of f(x), obtain an approximation for $\sqrt{3}$

Give your answer to 4 decimal places.

(2)

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(Total for question = 6 marks)

(Q04 WMA14/01, Oct 2021)

Q6.

$$g(x) = \frac{1}{\sqrt{4 - x^2}}$$

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(a) Find, in ascending powers of x, the first four non-zero terms of the binomial expansion of g(x). Give each coefficient in simplest form.

(5)

(b) State the range of values of x for which this expansion is valid.

(1)

(c) Use the expansion from part (a) to find a fully simplified rational approximation for $\sqrt{3}$

Show your working and make your method clear.

(2)

(Total for question = 8 marks)

(Q04 WMA14/01, Oct 2022)

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Q7.

(a) Find the first four terms, in ascending powers of x, of the binomial expansion of

$$\frac{8}{(2 - 5x)^2}$$

writing each term in simplest form.

(4)

(b) Find the range of values of x for which this expansion is valid.

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(1)

(Total for question = 5 marks)

(Q01 WMA14/01, Oct 2023)

Q8.

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(a) Find the first 4 terms of the binomial expansion, in ascending powers of x, of

$$\left(\frac{1}{4}-\frac{1}{2}x\right)^{-\frac{3}{2}} \qquad |x|<\frac{1}{2}$$

giving each term in simplest form.

(5)

Given that

$$\left(\frac{1}{4}-\frac{1}{2}x\right)^{n}\left(\frac{1}{4}-\frac{1}{2}x\right)^{-\frac{3}{2}}=\left(\frac{1}{4}-\frac{1}{2}x\right)^{\frac{1}{2}}$$

(b) write down the value of n.

(1)

(c) Hence, or otherwise, find the first 3 terms of the binomial expansion, in ascending powers of x, of

$$\left(\frac{1}{4}-\frac{1}{2}x\right)^{\frac{1}{2}} \qquad |x|<\frac{1}{2}$$

giving each term in simplest form.

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(3)

(Total for question = 9 marks)

(Q01 WMA14/01, June 2023)

Q9.

(a) Find, in ascending powers of x, the first three non-zero terms of the binomial series expansion of

$$\sqrt[3]{1 + 4x^3} \qquad |x| < \frac{1}{\sqrt[3]{4}}$$

giving each coefficient as a simplified fraction.

(4)

(b) Use the expansion from part (a) with $x = \frac{1}{3}$ to find a rational approximation to $\sqrt[3]{31}$

(3)

(Total for question = 7 marks)

(**Q02 WMA14/01, Jan 2022**)

Q10.

The binomial expansion of

$$(3 + kx)^{-2} \qquad |kx| < 3$$

where k is a non-zero constant, may be written in the form

$$A + Bx + Cx^2 + Dx^3 + \ldots$$

where A, B, C and D are constants.

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(a) Find the value of A

(1)

Given that $C = 3B$

(b) show that

$$k^2 + 6k = 0$$

(3)

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(c) Hence (i) find the value of k

(ii) find the value of D

(3)

(Total for question = 7 marks)

(Q01 WMA14/01, June 2022)

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